Lab 3 — Architecture Selection & Justification

Problem Statement #39 — Second-Hand Product Escrow Marketplace

Architecture Selection: We chose Microservices Architecture for the Second-Hand Product Escrow Marketplace System.

Architectural Choice

The system is decomposed into six independently deployable services — API Gateway, Order & Escrow Service, Inspection Service, Payment Service, Dispute Resolution Service, and Audit Log Service — each owning a single responsibility from the requirements table and communicating through well-defined provided/required interfaces (e.g., FundsControl, VerdictUpdate, AuditLogging).

Reason 1 — Independent Scaling & Fault Isolation for Critical Money Flows

FR-001, FR-003 and NFR-001 place hard, low-latency guarantees on escrow locking, dual-verification release, and dispute freezing. A microservices split lets the Payment Service and Order & Escrow Service be scaled and load-tested independently of lower-traffic services like Inspection or Audit Logging, and a failure or slowdown in one service (e.g., report/photo uploads in the Inspection Service) cannot block or delay the fund-freeze path enforced by NFR-001.

Reason 2 — Independent Deployability Matching Distinct Actor Workflows

Buyer-facing flows (place order, sign off, raise disputes) and Inspection-Agent-facing flows (submit verdicts, upload evidence) evolve on different release cadences and are owned by different sub-teams in practice. Microservices allow the Inspection Service to be updated, redeployed, or even reimplemented without redeploying the Order & Escrow or Payment services, reducing regression risk on the money-handling core.

Security Advantage

Isolating the Audit Log Service as its own microservice, with a single narrow AuditLogging provided interface and no direct database access from any other service, satisfies NFR-002's tamper-evident, append-only logging requirement: every escrow, payment, and dispute action is written through one controlled interface, so a compromise of any other service (e.g., Inspection) cannot be used to rewrite or delete audit history — the blast radius of a breach is contained to the compromised service's own data store.

Performance Benefit

Because the Payment Service and Audit Log Service are separate deployable units, they can each be given dedicated resources and horizontally scaled during peak load (e.g., flash-sale traffic), so NFR-001's target of an escrow freeze applied in under 500 ms under simulated peak load can be met without that latency budget being shared with — or degraded by — unrelated, bursty workloads such as report generation in the Inspection Service.

Trade-offs Considered

- Operational complexity (service discovery, distributed tracing, more deployment pipelines) is higher than a layered monolith — accepted because the security and latency guarantees around escrow funds are the dominant, non-negotiable requirement.
- Cross-service consistency (e.g., order state vs. escrow balance) requires careful interface contracts and idempotent calls between Order & Escrow, Payment, and Audit Log services, mitigated by the AuditLogging interface acting as the single source of truth for what actually happened.